

CL Synergy Freight

Sales Tracking System

MVP Documentation

Workflows, Roles & System Architecture

Phase 1: Ocean Freight Operations

Version 1.0 | MVP Demo

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1. Executive Summary

CL Synergy Freight's Sales Tracking System is a comprehensive web application designed to consolidate and streamline freight forwarding operations for CL Synergy, a Sri Lanka-based shipping and logistics company. The system replaces fragmented communication channels (WhatsApp, Email, Phone) with a unified platform that manages the complete lifecycle of shipping inquiries from initial customer contact through to final delivery.

Purpose

The MVP (Minimum Viable Product) focuses on Phase 1: Ocean Freight operations. It provides a single source of truth for all inquiry, quotation, booking, and shipment data while enforcing business rules around customer compliance (KYC), pricing margins, and role-based access control.

Key Capabilities

- End-to-end inquiry lifecycle management (11-stage workflow)
- AI-powered chat assistant with natural language processing for data entry and lookups
- Multi-line quotation builder with margin-based auto-approval routing
- Booking management with normal and urgent (spot rate) workflows
- Multi-leg shipment tracking with transshipment monitoring
- Role-based access control for 5 organizational roles
- KYC compliance tracking and enforcement
- Real-time dashboard with KPIs, hit rates, and operational analytics
- Customer blacklist and credit hold enforcement across all modules
- Audit trail and activity logging for all operations

Technology Stack

- Frontend: React 18 + TypeScript + Vite (with Recharts for analytics)
- Backend: Python FastAPI with WebSocket support
- AI Integration: OpenAI GPT with 40+ domain-specific tool definitions
- Data Storage: JSON file-based persistence with atomic writes (MVP phase)

2. System Overview

Application Structure

The system is organized as a modular monolith with a React single-page application (SPA) frontend communicating with a Python FastAPI backend. The frontend handles UI rendering, local state management, and optimistic updates, while the backend provides REST API endpoints, AI chat processing via WebSocket, and persistent data storage.

Frontend Modules (9 Pages)

1. Dashboard - Operations snapshot with KPIs, charts, and recent activity
2. Command Center (Chat) - AI-powered chat with structured quick commands
3. Inquiry List - Tabular view of all inquiries with filters and workflow management
4. Quotations - Quote list, builder modal, and approval workflow
5. Shipments - Multi-leg tracking with timeline visualization
6. Operations (Follow-ups) - Follow-up logging, task management, missing items
7. Customers - Customer master data with tier management and compliance flags
8. KYC Form - Know Your Customer compliance onboarding
9. Workspace - Unified operations hub with role-specific workflow steps

Backend Modules

- REST API Layer (frontend_api.py): 25+ endpoints for CRUD operations
- AI Chat Engine (init_chat.py): OpenAI GPT integration with tool calling
- Tool Definitions (tool_definitions.py): 40+ function schemas for AI operations
- Tool Handler (tool_handler.py): Dispatches AI tool calls to database operations
- Database Layer (queries.py): Generic CRUD with atomic JSON file persistence
- Frontend Query Layer (frontend_queries.py): ID mapping and data transformation

Data Flow Architecture

All state is centralized in the App.tsx component on the frontend, using a custom usePersistentState hook that syncs to localStorage. Mutations follow an optimistic update pattern: the UI updates immediately, then an API call is sent to the backend in the background. The backend persists all data to a single JSON file (mock_data.json) using thread-safe atomic writes. On page load, the frontend fetches fresh data from the backend via the /api/dashboard/init endpoint.

3. User Roles & Permissions

The system implements Role-Based Access Control (RBAC) with 5 distinct organizational roles. Each role has specific page access permissions and action permissions. The active role is determined by the currently selected employee in the top bar.

3.1 Customer Service (CS)

The primary operational role responsible for customer interaction, inquiry intake, and day-to-day workflow management. CS agents are the first point of contact for all customer communications.

Page Access:

- Dashboard, Chat, Workspace, Inquiry List, Operations, Customers, KYC

Key Responsibilities:

- Create and manage customer inquiries from all channels
- Perform customer verification (blacklist/credit hold checks)
- Send KYC forms to new customers
- Log follow-ups and manage tasks
- Send approved quotations to customers
- Create bookings from confirmed quotes
- Release bookings to customers
- Advance inquiries through workflow stages

3.2 Sales Executive

Responsible for revenue generation through quotation preparation, customer relationship management, and inquiry conversion tracking.

Page Access:

- Dashboard, Chat, Workspace, Inquiry List, Quotations, Operations, Customers

Key Responsibilities:

- Prepare multi-line quotations with shipping line options
- Manage customer follow-ups and conversion pipeline
- Track hit rates (inquiry-to-completion conversion)
- Create new inquiries from customer interactions

3.3 Finance

Responsible for financial compliance, customer vetting, and quotation approval for pricing that falls below margin thresholds.

Page Access:

- Dashboard, Chat, Workspace, Customers, KYC, Quotations

Key Responsibilities:

- Verify and approve/reject KYC submissions
- Approve or reject quotations below minimum margin thresholds
- Manage customer blacklist and credit hold flags
- Adjust customer tier classifications and payment terms

3.4 Procurement

Responsible for sourcing shipping rates from carriers, confirming vessel space, and managing liner relationships.

Page Access:

- Dashboard, Chat, Workspace, Inquiry List, Operations

Key Responsibilities:

- Source rates from shipping lines (AMS system and INTTRA)
- Confirm booking space with liners (vessel and voyage assignment)
- Respond to urgent/spot rate requests within 15-minute windows
- Manage rate database and carrier contracts

3.5 Admin

Full system access for supervisory oversight, configuration, and troubleshooting. Admin has unrestricted access to all pages and all commands.

Page Access: All pages

Permissions: All actions available

3.6 Default Employee Roster

ID	Name	Role
1	Nimal Perera	Sales Executive
2	Anjali Silva	Customer Service (Default)
3	Rohan Fernando	Sales Executive
4	Priya Jayawardena	Finance
5	Kamal Dissanayake	Procurement

4. Core Data Model

The system manages several interconnected entities that represent the freight forwarding business domain. Below are the primary entities and their key attributes.

4.1 Customer

Represents a shipping customer or client. Customers are classified by tier which determines minimum pricing margins, payment terms, and service levels.

- Fields: id, name, location, tier, payment_terms, contact_email, contact_phone
- Flags: blacklisted (blocks all operations), credit_hold (warns but allows)
- Compliance: kyc_status (not_started / pending_customer / approved / rejected)
- Pricing: min_margin_pct (tier-based floor for quotation margins)

4.2 Inquiry

Represents a customer request for shipping services. Each inquiry tracks the full journey from initial contact through to completion.

- Fields: id, customer_name, inquiry_text, request, origin, destination
- Classification: delivery_type, channel (WhatsApp/Email/Phone), sbu (business unit)
- Tracking: status (pending/completed), workflow_stage (11 stages), employee_id
- Timestamps: created_at, completed_at

4.3 Quote

A pricing proposal sent to a customer. Quotes contain multiple rate line options from different shipping carriers, allowing the customer to choose their preferred option.

- Fields: id, inquiry_id, customer_name, origin, destination, quote_type
- Pricing: margin_pct, lines[] (array of QuoteLine objects)
- Status: Draft > Awaiting Approval > Approved > Sent > Confirmed / Lost
- Approval: approval_reason (populated when margin below threshold)

4.4 Quote Line

An individual shipping option within a quote, representing a specific carrier's offering.

- Fields: shipping_line, rate_type, base_rate_usd, transit_days
- Details: free_time_days, transshipment_points, destination_charges_usd

4.5 Booking

Represents a confirmed shipping reservation with a carrier. Bookings track the complete container lifecycle from reservation through to release.

- Fields: id, quote_id, customer_name, origin, destination, shipping_line
- Vessel: vessel_name, voyage_number, container_type, quantity
- Status: Pending Liner > Liner Confirmed > Released / Cancelled
- Documents: SI cutoff/requested/submitted, BL cutoff, draft BL, master BL, house BL
- Flags: is_urgent (spot rate bypass)

4.6 Shipment

Tracks the physical movement of goods from origin to destination. Shipments contain multiple legs representing each port of call.

- Fields: id, quote_id, customer_name, origin, destination, shipping_line
- Status: Booked > In Transit > At Transshipment > Out for Delivery > Delivered
- Tracking: legs[] (array of ShipmentLeg), expected_delivery, pod_received

4.7 Shipment Leg

An individual segment of a shipment's journey (e.g., Origin Port, Transshipment Port, Destination Port).

- Fields: port, type (Origin/Transshipment/Destination)
- Tracking: expected_at, actual_at, status (Pending/Arrived/Departed/Delayed)

4.8 Supporting Entities

Follow-up

- Notes and interactions logged against an inquiry (id, inquiry_id, note, employee_id)
- Optional completion_flag to mark the linked inquiry as completed

Task

- Action items and reminders (id, customer_name, task, status, due_date, employee_id)

Missing Item

- Outstanding documents blocking progress (id, customer_name, missing_item, cutoff_date)
- Cutoff date enforcement: items turn red in UI when overdue

Activity Entry

- Audit log of all system actions (actor_role, action, ref_type, ref_id, timestamp)

Rate Record (AMS)

- Internal shipping rate database (liner, route, container_type, rate, validity period)

5. Primary Workflow: Inquiry Lifecycle

The inquiry lifecycle is the core workflow of the system, consisting of 11 sequential stages that an inquiry passes through from initial receipt to completion. Each stage is associated with a responsible role and specific actions that must be completed before advancing to the next stage.

5.1 Workflow Stages

Stage 1: Inquiry Received (Owner: CS)

A customer submits a shipping inquiry via WhatsApp, Email, or Phone. CS logs the inquiry in the system with route details, container requirements, delivery type, and SBU classification.

Stage 2: Customer Verification (Owner: CS)

CS verifies the customer's status. Blacklisted customers are blocked (inquiry cannot proceed). Customers on credit hold receive a warning but may proceed with confirmation.

Stage 3: KYC Initialization (Owner: CS)

For new customers without approved KYC, CS sends the KYC form via the KYC page. The customer's KYC status changes from 'not_started' to 'pending_customer'.

Stage 4: KYC Clearance (Owner: Finance)

Finance reviews the submitted KYC documents and either approves or rejects the customer. Only approved customers can proceed further in the workflow.

Stage 5: Rate Check (AMS) (Owner: CS)

CS checks the internal rate management system (AMS) for available shipping rates on the requested route. If rates exist, they proceed to quotation. If not, a procurement request is raised.

Stage 6: Procurement Request (Owner: Procurement)

Procurement contacts shipping lines to obtain rates. This includes checking INTTRA for spot rates and negotiating with carriers. For urgent/spot requests, a 15-minute confirmation window applies.

Stage 7: Quotation Preparation (Owner: Sales)

Sales builds a multi-line quotation using rates from AMS or Procurement. Each line represents a different shipping option (carrier, rate, transit time). Margin percentage is applied and validated against the customer's minimum threshold.

Stage 8: Quote Sent (Owner: CS)

CS sends the approved quotation to the customer via their preferred communication channel. The system simulates email delivery with a toast notification.

Stage 9: Customer Response (Owner: CS)

The system awaits the customer's decision. CS logs follow-ups during this waiting period. The customer either confirms (accepts a specific line) or the quote is marked as lost.

Stage 10: Booking Request (Owner: CS)

Once a quote is confirmed, CS creates a booking from the selected shipping line option. Normal bookings go to Procurement for liner confirmation; urgent bookings skip directly to confirmed.

Stage 11: Completed (Owner: CS)

The inquiry is marked as completed once the shipment is confirmed and tracking has begun. The completion timestamp is recorded and the inquiry contributes to hit rate metrics.

5.2 Workflow Advancement

Inquiries can be advanced through stages via two mechanisms:

- Inquiry List Page: Each inquiry row has a WorkflowStepper component that displays the current stage and allows manual advancement to the next stage via a click action.
- Workspace Page: The unified operations hub presents role-specific steps with action modals. Users complete the required action in a modal form, and the workflow stage advances automatically.

5.3 Stage Visualization

The WorkflowStepper component renders the 11 stages as a horizontal or vertical progress indicator. Completed stages show a green checkmark, the current stage is highlighted in blue, and future stages appear in grey. The component is interactive, allowing users to click on the next stage to trigger advancement.

6. Quotation Workflow

The quotation workflow manages the creation, approval, and customer communication of pricing proposals. It includes an automated approval routing mechanism based on margin thresholds.

6.1 Quote Creation

Quotations are created from the Quotations page via the 'New Quote' button or from the Workspace. The quote builder modal provides:

- Customer selection with autocomplete (displays tier, payment terms, minimum margin)
- Route configuration (origin, destination, quote type: FCA/Domestic/Drayage/DDP)
- Rate line builder for adding multiple shipping line options
- Margin percentage input with real-time validation against customer floor

6.2 Multi-Line Quoting

Each quote can contain multiple rate lines, where each line represents a different shipping carrier option. This allows customers to compare options across carriers. Each line includes:

- Shipping line (Maersk, CMA CGM, MSC, Hapag-Lloyd, ONE, Evergreen, Yang Ming, ZIM)
- Rate type: Spot (instant, 15-min validity), Contractual (pre-negotiated), NAC (named account), Convoy (group)
- Pricing: Base rate (USD), destination charges (USD)
- Service: Transit days, free time (detention/demurrage days), transshipment points

6.3 Quote Type Classification

- FCA (Free Carrier): Ocean freight only; customer arranges land transport on both ends
- Domestic Included: Ocean freight + local pickup at origin
- Drayage: Ocean freight + destination haulage to final location
- DDP (Delivered Duty Paid): Full door-to-door service including customs clearance

6.4 Margin-Based Approval Routing

When a quote is saved, the system compares the applied margin percentage against the customer's minimum margin floor (determined by their tier). The routing logic is:

- If margin $\geq$ customer's min_margin_pct: Quote saved as 'Draft' (can be sent directly)
- If margin $<$ customer's min_margin_pct: Quote auto-routed to 'Awaiting Approval' status
- Approval reason recorded (e.g., 'margin 3% $<$ min 5%') for audit trail
- Finance must approve before quote can be sent to customer

6.5 Quote Status Transitions

Quotations progress through the following status chain:

1. Draft - Initial state after creation (above-threshold margin)

2. Awaiting Approval - Auto-set when margin below threshold; requires Finance action
3. Approved - Finance has approved the below-threshold quote
4. Sent - Quote has been sent to the customer (email simulation)
5. Confirmed - Customer has accepted the quote (triggers booking creation)
6. Lost - Customer declined or did not respond within the valid period

6.6 Customer Tier Margin Thresholds

Tier	Min Margin %	Typical Payment Terms
Key Account	5%	60-Day Credit
Regular	7%	30-Day Credit
Walk-in	10%	Pay Upfront

7. Booking Workflow

The booking workflow manages the reservation of shipping container space with carriers. Two distinct paths exist: a normal flow that involves Procurement confirmation and an urgent (spot rate) flow that bypasses Procurement.

7.1 Normal Booking Flow

1. CS creates booking from a confirmed quote, specifying shipping line, container type, and quantity. Status: 'Pending Liner'.
2. Procurement is notified and contacts the shipping line to confirm vessel space.
3. Procurement confirms the booking with vessel name and voyage number. Status: 'Liner Confirmed'.
4. CS releases the booking details to the customer. Status: 'Released'.
5. Shipment tracking begins once the booking is released.

7.2 Urgent (Spot Rate) Booking Flow

For spot rate inquiries requiring immediate action (15-minute confirmation window), the urgent flow bypasses the Procurement queue:

1. CS creates booking with is_urgent=true flag. Status set directly to 'Liner Confirmed'.
2. CS notifies Procurement after the fact (acknowledgment required).
3. CS releases the booking to the customer. Status: 'Released'.

7.3 Document Workflow (SI & BL)

After booking confirmation, the system tracks critical shipping documents with cutoff dates:

Shipping Instructions (SI)

- SI cutoff date set by CS
- SI requested from customer (flag)
- SI submitted to INTTRA/carrier (flag)

Bill of Lading (BL)

- BL cutoff date recorded
- Draft BL sent for review (flag)
- Master BL number assigned (from carrier)
- House BL number created (by CL Synergy)
- BL approval status: pending / approved / changes-requested

7.4 Booking Status Summary

Status	Description	Next Action
Pending Liner	Awaiting Procurement	Confirm with carrier
Liner Confirmed	Vessel space secured	Release to customer

Released

Customer notified

Begin tracking

Cancelled

Booking cancelled

N/A

8. Shipment Tracking Workflow

Shipment tracking provides visibility into the physical movement of goods from origin to destination. The system supports multi-leg journeys with independent tracking per leg, addressing a critical operational gap identified for transshipment monitoring.

8.1 Shipment Statuses

- Booked: Vessel space confirmed, awaiting departure
- In Transit: Shipment has departed origin, moving between ports
- At Transshipment: Goods at an intermediate port (high-risk for delays)
- Out for Delivery: Final leg in progress (road/yard delivery)
- Delivered: Proof of Delivery (POD) received; company no longer liable
- Delayed: One or more legs have missed expected timelines

8.2 Multi-Leg Tracking

Each shipment contains an ordered array of legs representing the journey segments. Legs are typed as Origin, Transshipment, or Destination. Each leg is independently tracked with expected and actual arrival/departure times.

Leg status progression: Pending > Arrived > Departed (destination legs stop at Arrived). Users advance leg status by clicking the advance button on the Shipments page timeline.

8.3 Automatic Status Computation

The shipment's overall status is automatically computed from its leg states:

- All legs pending or origin departed: 'In Transit'
- Any transshipment leg arrived: 'At Transshipment'
- Destination leg arrived: 'Delivered'
- Any leg with 'Delayed' status: 'Delayed' override

8.4 Critical Watch - Transshipment Monitoring

The Shipments page prominently displays a 'Critical Watch' banner highlighting all shipments currently at transshipment ports. This addresses a key operational concern where goods can be delayed or misrouted at intermediate ports, requiring immediate attention from the operations team.

8.5 Proof of Delivery (POD)

Once a shipment's destination leg is reached, CS can record the POD date. This action marks the shipment as 'Delivered', triggers a simulated email notification to the customer, and completes the shipping cycle.

9. KYC Compliance Workflow

The Know Your Customer (KYC) workflow ensures regulatory compliance by requiring customer identity verification before significant business transactions can proceed.

9.1 KYC Status Lifecycle

1. Not Started: New customer registered, no KYC form sent yet
2. Pending Customer: KYC form sent, awaiting customer's document submission
3. Approved: Finance has reviewed and approved the KYC documents
4. Rejected: Finance has found issues with submitted documents

9.2 KYC Process Steps

- CS creates new customer (KYC status defaults to 'not_started')
- CS navigates to KYC Form page and sends KYC form to customer
- System updates customer's KYC status to 'pending_customer'
- Customer submits required documents (simulated in MVP)
- Finance reviews documents on KYC page
- Finance approves or rejects the KYC submission
- Inquiry workflow auto-advances past KYC stages for approved customers

9.3 KYC Enforcement

The inquiry workflow stages 3 (KYC Initialization) and 4 (KYC Clearance) enforce KYC compliance. Inquiries for customers without approved KYC cannot advance past these stages until the KYC process is completed. Existing customers with approved KYC bypass these stages automatically.

10. Customer Management

10.1 Customer Tiers

Customers are classified into three tiers that determine pricing and service levels:

- Key Account (5% min margin, 60-day credit): High-volume, long-term strategic customers
- Regular (7% min margin, 30-day credit): Standard recurring customers with established history
- Walk-in (10% min margin, Pay Upfront): One-off or new customers without credit history

10.2 Blacklist & Credit Hold

Blacklist

Blacklisted customers are completely blocked from all system operations. The AI chat refuses to create inquiries for blacklisted customers and displays a red error bubble. On the Customers page, blacklisted customers appear at the bottom with a red tint and shield-alert badge.

Credit Hold

Customers on credit hold receive a yellow warning. The system allows operations to proceed after explicit confirmation ('Proceed anyway?'). These customers appear with a yellow tint and warning badge on the Customers page.

10.3 Customer Auto-Creation

When the AI chat assistant receives an inquiry referencing a customer not in the system, it automatically offers to create the customer. The AI asks whether the customer is Recurring (Regular tier) or One-Time (Walk-in tier) and creates the record accordingly.

11. AI Chat Assistant (Command Center)

The AI Chat Assistant is a natural language interface powered by OpenAI GPT that allows users to perform system operations through conversational commands. It connects via WebSocket for real-time communication and has access to 40+ domain-specific tools.

11.1 Quick Commands

The chat provides structured forms for common operations (availability varies by role):

- /new customer: Register a new customer with tier, location, and payment terms
- /new inquiry: Log a shipping inquiry with route, container details, channel, and SBU
- /follow up: Log a follow-up note against an inquiry (optionally mark complete)
- /new task: Create a task or reminder assigned to an employee
- /quote: Open the quotation builder for a specific customer
- /lookup: Search customers, inquiries, or rates by various criteria
- /blacklist: Toggle a customer's blacklist flag
- /credit hold: Toggle a customer's credit hold flag
- /change tier: Promote or demote a customer's tier classification
- /complete: Mark an inquiry as completed
- /new booking: Create a booking from a confirmed quotation
- /confirm booking: Procurement confirms liner space with vessel details
- /release booking: CS releases container instructions to the customer

11.2 Natural Language Processing

Beyond structured commands, the AI understands free-form natural language. For example, typing 'Hayleys needs 12 reefers from Colombo to Hamburg' will be parsed into a structured inquiry with the customer, container count, origin, and destination extracted automatically.

11.3 Business Rule Enforcement

The AI enforces business rules during chat interactions:

- Blocks operations for blacklisted customers (red error bubble)
- Warns about credit hold customers (yellow warning bubble with confirmation)
- Handles customer name ambiguity (asks to select from similar matches)
- Auto-creates new customers when referenced but not found in the system
- Deduplicates customer names to prevent duplicate records

11.4 Tool-Based Architecture

The AI uses OpenAI's function calling feature with 40+ tool definitions covering all CRUD operations. When the AI determines an action is needed, it calls the appropriate tool, the backend executes the operation on the database, and

the result is fed back to the AI for a natural language confirmation response.

12. Dashboard & Analytics

The Dashboard provides an at-a-glance operational snapshot with key performance indicators, charts, and activity feeds.

12.1 KPI Cards

- Pending Inquiries: Count of open inquiries requiring action
- Completed Today: Inquiries resolved in the current day
- Active Customers: Customers with at least one inquiry in the system
- Overdue Items: Missing documents past their cutoff date
- Pending Approvals: Quotations awaiting Finance approval

12.2 Lifetime Metrics

- Total Inquiries: All-time inquiry count
- Completed Inquiries: All-time completions
- Completion Rate: Percentage of inquiries successfully completed

12.3 Charts & Visualizations

- Salesperson Hit Rate: Bar chart showing inquiry-to-completion conversion per employee
- SBU Breakdown: Distribution of inquiries across Strategic Business Units
- Inquiries Per Day: Time series chart showing pending vs completed inquiry trends
- Channel Distribution: Pie chart showing WhatsApp/Email/Phone inquiry proportions

12.4 Natural Language Query Box

The Dashboard includes an NLQ (Natural Language Query) search box that allows users to type free-form questions about the data, which are processed by the AI backend.

12.5 Recent Activity Feed

Displays the 5 most recent inquiries with customer name, route, status badge, and timestamp, providing a quick overview of current operational activity.

13. Workspace (Unified Operations Hub)

The Workspace page serves as a unified operations hub that organizes all pending work items by role-specific workflow steps. It presents an inbox-style interface where each item maps to a specific action that must be completed.

13.1 Role-Specific Steps

- Customer Service: 14 steps including inquiry intake, customer checks, KYC, quoting, booking, SI/BL tracking
- Finance: 2 steps - KYC approval and quote approval
- Procurement: 3 steps - rate sourcing, booking confirmation, urgent alert handling
- Sales: 1 step - quotation preparation
- Admin: 20 steps - all steps from all roles combined

13.2 Inbox Organization

The inbox organizes pending items by type: Inquiries, Bookings, Quotes, and Customers. Each item displays its current workflow stage, associated customer, and available actions. Users click on an item to open a modal with the specific action form for that step.

13.3 Action Modals

The Workspace provides specialized modals for each workflow action:

- Advance inquiry workflow stage
- Send KYC form to customer
- Check rates in AMS or INTTRA
- Prepare quotation with rate lines
- Confirm booking (vessel and voyage assignment)
- Release booking to customer
- Record SI cutoff dates and submission
- Record Master BL and House BL numbers
- Set BL approval status

13.4 Activity Log

The Workspace maintains a timeline of all actions performed by the current role, providing a complete audit trail of operational activities.

14. Business Rules & Logic

14.1 Margin Enforcement

Every quotation's margin percentage is validated against the customer's tier-based minimum. Key Account customers require at least 5%, Regular at least 7%, and Walk-in at least 10%. Quotes below these thresholds are automatically routed to Finance for approval before they can be sent to the customer.

14.2 Blacklist Blocking

Blacklisted customers are completely blocked from all transactional operations. The AI chat refuses to create inquiries, follow-ups, or bookings. The system displays red indicators across all pages where the customer appears.

14.3 Credit Hold Warnings

Customers on credit hold trigger yellow warning indicators. Operations can proceed after explicit user confirmation, but the warning ensures awareness of the financial risk.

14.4 Urgent Inquiry Detection

The system scans inquiry text for urgency keywords ('urgent', 'spot', 'asap', 'same-day', 'immediate'). Flagged inquiries appear in a prominent 'Spot Rate / Urgent - Time Critical' banner on the Operations page with a 15-minute confirmation window notice.

14.5 Most-Chased Inquiries Metric

The Operations page ranks pending inquiries by follow-up count to identify bottlenecks. Inquiries with many follow-ups indicate Procurement delays or customer indecision. Color coding: green (0-5 follow-ups), amber (5-10), red (10+).

14.6 Payment Terms by Tier

- Key Account: 60-Day Credit (maximum float)
- Regular: 30-Day Credit (standard terms)
- Walk-in: Pay Upfront (no credit risk)

15. Technical Architecture

15.1 Frontend Architecture

The frontend is a React 18 single-page application built with TypeScript and Vite. All application state is centralized in the App.tsx component using the custom usePersistentState hook, which syncs state to localStorage for persistence across browser refreshes.

- Framework: React 18 with functional components and hooks
- Build Tool: Vite for fast development and optimized production builds
- Language: TypeScript for type safety
- Charts: Recharts for data visualization
- Icons: Lucide-react icon library
- State: Centralized in App.tsx with localStorage persistence
- Permissions: React Context API for role-based access control
- API Client: Fetch-based REST client with optimistic updates
- Real-time: WebSocket client for AI chat communication

15.2 Backend Architecture

The backend is a Python FastAPI application providing both REST API endpoints and WebSocket connectivity. It uses a JSON file as the data store with atomic write operations for data integrity.

- Framework: FastAPI (async HTTP + WebSocket)
- AI Integration: OpenAI API with function/tool calling
- Data Store: JSON file (mock_data.json) with thread-safe atomic writes
- Architecture: Modular monolith (Routes, Chat, DB, Tools in separate modules)
- ID Mapping: Backend numeric IDs transformed to frontend string IDs (CUS-001, INQ-1041)
- CORS: Custom middleware wrapper for WebSocket compatibility

15.3 AI Integration Architecture

The AI system uses OpenAI's function calling feature with a domain-specific system prompt and 40+ tool definitions. When a user sends a message via WebSocket, the backend forwards it to OpenAI with conversation history and available tools. OpenAI may respond with tool calls, which are executed against the database, with results fed back for a natural language response.

15.4 Data Persistence

The MVP uses a dual-persistence strategy: the frontend stores state in localStorage for immediate access, while the backend persists to a JSON file using atomic writes (temp file + os.replace) with thread locking to prevent data corruption from concurrent access.

16. API Reference

All REST API endpoints are prefixed with /api and served by the FastAPI backend.

16.1 Dashboard

- GET /api/dashboard/init - Bulk load all data (customers, inquiries, tasks, quotes, shipments, bookings, etc.)

16.2 Inquiries

- POST /api/inquiries - Create a new inquiry
- POST /api/inquiries/{id}/complete - Mark inquiry as completed
- POST /api/inquiries/reopen - Reopen a completed inquiry
- PATCH /api/inquiries/{id}/workflow-stage - Advance to next workflow stage

16.3 Customers

- POST /api/customers - Create a new customer
- PATCH /api/customers/{name} - Update customer attributes

16.4 Quotations

- POST /api/quotes - Create a new quotation
- PATCH /api/quotes/{id}/status - Update quote status (transition)

16.5 Bookings

- POST /api/bookings - Create a new booking
- PATCH /api/bookings/{id}/confirm - Confirm liner space (vessel + voyage)
- PATCH /api/bookings/{id}/release - Release booking to customer
- POST /api/bookings/{id}/notify-procurement - Mark procurement notified (urgent)
- PATCH /api/bookings/{id}/si-cutoff - Set SI cutoff date
- PATCH /api/bookings/{id}/si-requested - Mark SI requested from customer
- POST /api/bookings/{id}/si-submit - Submit SI to carrier
- PATCH /api/bookings/{id}/bl-cutoff - Set BL cutoff date
- PATCH /api/bookings/{id}/master-bl - Record master BL number
- POST /api/bookings/{id}/house-bl - Create house BL

16.6 Shipments

- PATCH /api/shipments/{id}/legs/{leg-id}/advance - Advance shipment leg status
- POST /api/shipments/{id}/pod - Record Proof of Delivery

16.7 Follow-ups & Tasks

- POST /api/followups - Create a follow-up note
- POST /api/tasks - Create a new task
- POST /api/tasks/{id}/complete - Mark task as completed

16.8 Activities & Rates

- POST /api/activities - Log an activity entry
- POST /api/rates/search - Search AMS rate database
- GET /api/rates/intra - Query INTTRA spot rates (stubbed)

16.9 WebSocket

- WS /ws - AI chat endpoint (bidirectional message stream)

17. Deployment & Configuration

17.1 Development Setup

Backend

Navigate to the BackEnd directory, create a Python virtual environment, install dependencies from requirements.txt, and run main.py. The server starts on `http://localhost:8000`.

Frontend

Navigate to FrontEnd/dashinterface, run `npm install`, then `npm run dev`. The development server starts on `http://localhost:5173` with hot module replacement.

17.2 Environment Variables

- OPENAI_API: OpenAI API key for GPT chat integration
- OPEN_AI_MODEL: Model identifier (gpt-4-turbo or gpt-4o-mini)
- RESEND_API: Email service API key (for sending real emails in production)
- VITE_API_BASE: Backend REST endpoint prefix (default: /api)
- VITE_WS_URL: WebSocket URL (default: ws://localhost:8000/ws)

17.3 Production Deployment

A Dockerfile is provided at the project root for containerized deployment. The frontend is built to static files and served from the /static directory via FastAPI, creating a single-container deployment. The application can be deployed to any Docker-compatible hosting platform.

17.4 Demo Reset

The 'Reset Demo' button in the top bar clears all localStorage data and resets the application to its seed data state. This is useful for demonstrations and testing.

18. Phase 2 Roadmap

The following features are planned for Phase 2 development, building upon the foundation established in the MVP.

18.1 Additional SBU Workflows

- Air Freight: Dedicated workflow for air cargo operations
- Domestic: Local Sri Lankan logistics and warehousing workflows

18.2 External Integrations

- INTTRA/Freightify: Live API integration for real-time spot rates and booking
- Big Schedule: Live carrier schedule data integration
- Email Service: Replace simulated toasts with actual email delivery (Resend API)
- Unified Inbox: WhatsApp, Email, Viber, WeChat message aggregation

18.3 Customer Portal

A customer-facing self-service portal where customers can submit inquiries, track shipments, download documents, and manage their profile. Currently represented as a placeholder banner on the Customers page.

18.4 Database Migration

Migration from JSON file-based storage to a proper relational database (PostgreSQL or similar) for production scalability, concurrent access, and data integrity.

18.5 Advanced Analytics

- Hit rate per SBU (currently only per salesperson)
- Revenue and profitability tracking
- Carrier performance scoring
- Customer lifetime value analysis

18.6 Other Planned Features

- Multi-language support for international operations
- Spot-rate live 15-minute countdown timer
- Complete audit log for all chat operations
- Multi-action chat commands (currently limited to one action per message)
- Real-time notifications and alerts system